

F1 Race Strategy & Resource Optimization System

This document contains the complete professional workflow, architecture, feature breakdown, technology stack, tools, and implementation plan for the F1-inspired Race Strategy & Resource Optimization System. The purpose of this project is to simulate how Formula 1 race engineers analyze telemetry data and make race strategy decisions during a race weekend.

1. Core Project Idea

The project is an interactive motorsport strategy simulation platform that uses simulated telemetry data to optimize race decisions such as: Tyre management Pit stop timing Fuel optimization ERS/Battery deployment Traffic window analysis Safety Car strategy adjustments The system behaves like a digital race engineer assistant rather than a generic AI dashboard.

2. Overall Project Workflow

Step 1: Backend generates simulated telemetry data.
Step 2: Data is sent in real-time using WebSockets.
Step 3: Frontend dashboard visualizes telemetry and race analytics.
Step 4: Strategy engine analyzes tyre wear, traffic, fuel, and ERS usage.
Step 5: AI-assisted recommendation system generates race strategy suggestions.
Step 6: User tests race scenarios using the What-If Simulator.
Step 7: Reports and summaries are generated using Docling.

3. Live Telemetry Dashboard

The dashboard acts as the central race command center. It continuously displays race telemetry and strategy data.

What This Feature Includes:

- Lap number and sector timing
- Tyre condition and wear percentage
- Fuel level tracking
- ERS/Battery percentage
- Current pace trends
- Driver gap timings
- Position tracking

Technologies Used:

- React
- Tailwind CSS
- Recharts
- WebSockets / Socket.IO

Tools Used:

- VS Code
- Figma
- GitHub

4. Tyre Degradation Engine

This module predicts tyre wear and overheating using race pace, tyre compound, and lap count.

What This Feature Includes:

- Tyre wear prediction
- Grip loss estimation
- Overheating detection
- Tyre life estimation
- Compound comparison

Technologies Used:

- Python
- FastAPI
- Predictive formulas
- JSON telemetry simulation

Tools Used:

- IBM Granite
- Langflow
- GitHub

5. Smart Pit Stop Strategy

This feature calculates the ideal pit stop timing based on tyre life, race pace, traffic, and fuel conditions.

What This Feature Includes:

- Best pit window calculation
- Undercut and overcut analysis
- Tyre recommendation
- Pit loss estimation
- Race gain prediction

Technologies Used:

- FastAPI / Node.js
- Python logic engine

- Telemetry simulation

Tools Used:

- Langflow
- Postman
- GitHub

6. Traffic Window Analysis

This module predicts traffic conditions after a pit stop and identifies clean-air opportunities.

What This Feature Includes:

- Clean-air prediction
- Rejoin position estimation
- Traffic cluster detection
- Gap timing analysis

Technologies Used:

- React visualizations
- Python timing calculations
- WebSockets

Tools Used:

- Recharts
- Figma
- VS Code

7. Safety Car Strategy Recalculation

This module dynamically recalculates race strategy when a Safety Car scenario is triggered.

What This Feature Includes:

- Instant strategy updates
- Pit advantage analysis
- Position reshuffling
- Strategy comparison during SC

Technologies Used:

- Simulation engine
- FastAPI
- WebSockets

Tools Used:

- Langflow
- GitHub

8. ERS / Battery Optimization

This feature analyzes battery deployment and harvesting efficiency.

What This Feature Includes:

- ERS deployment prediction
- Overtake boost timing
- Energy harvesting analysis
- Battery efficiency tracking

Technologies Used:

- Python
- Granite AI assistance
- Telemetry simulation

Tools Used:

- IBM Granite
- Langflow

9. Fuel Usage Optimization

This module compares aggressive and balanced fuel strategies.

What This Feature Includes:

- Fuel consumption analysis
- Lift-and-coast prediction
- Fuel-saving suggestions
- Aggressive vs balanced modes

Technologies Used:

- FastAPI
- Python analytics
- Simulation engine

Tools Used:

- IBM Granite
- Postman

10. Strategy Comparison Engine

The system compares multiple racing strategies and predicts possible outcomes.

What This Feature Includes:

- Aggressive strategy simulation
- Balanced strategy simulation
- Tyre-saving strategy simulation
- Estimated finishing position

Technologies Used:

- Python prediction logic
- React visualizations

Tools Used:

- Granite AI
- Recharts

11. What-If Race Simulator

This is the most interactive feature of the project. Users can test different race scenarios.

What This Feature Includes:

- Safety Car simulation
- Pit timing experiments
- Tyre compound changes
- Fuel mode comparison
- Aggressive driving mode simulation

Technologies Used:

- Simulation engine
- WebSockets
- Frontend interaction system

Tools Used:

- React
- Langflow
- GitHub

12. AI Integration Layer

IBM AI tools will be used to improve recommendation quality and structured strategy insights.

What This Feature Includes:

- Telemetry interpretation

- Strategy recommendation summaries
- Intelligent analysis generation
- Race insight generation

Technologies Used:

- IBM Granite
- Langflow

Tools Used:

- IBM Learning Lab
- GitHub

13. Documentation & Reporting

The system will generate structured strategy reports and race summaries.

What This Feature Includes:

- Race summary generation
- Strategy reports
- Telemetry logs
- AI-generated insights

Technologies Used:

- Docling
- PDF generation

Tools Used:

- IBM Docling

14. Team Workflow & Responsibilities

Frontend Team Responsibilities: Dashboard UI Charts and visualizations Real-time telemetry display What-If Simulator interface **Backend Team Responsibilities:** Telemetry generation Strategy engine logic API development Real-time communication **AI & Analytics Responsibilities:** Granite integration Prediction systems Telemetry analysis Strategy recommendation generation **Documentation & Testing:** GitHub management Docling reporting Testing and debugging Presentation preparation

15. Final Project Goal

This project is designed as a realistic motorsport decision-support and strategy simulation platform. The focus is on interactive telemetry analysis, predictive race strategy, and practical AI-assisted optimization instead of unrealistic futuristic claims. The final goal is to create a professional race engineering simulator experience with strong visualization, intelligent strategy suggestions, and smooth real-time interactions.